

## EDUCATION

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### Massachusetts Institute of Technology

Cambridge, MA

*Ph.D. candidate in Electrical Engineering and Computer Science, GPA: 5.0/5.0*

*Fall 2023 – Expected Spring 2028*

- **Advisor:** Anantha P. Chandrakasan (Energy-Efficient Circuits and Systems Group)
- **Focus:** Hardware Security, Cryptographic ASIC Design

### Seoul National University

Seoul, South Korea

*M.S. in Intelligence and Information, GPA: 4.0/4.0*

*Fall 2021 – Summer 2023*

- **Advisor:** Dongsuk Jeon (Mobile Multimedia Systems Group)
- **Focus:** Hardware Security, FHE Accelerator

### Seoul National University

Seoul, South Korea

*B.S. in Electrical and Computer Engineering, Summa Cum Laude, GPA: 3.92/4.0*

*Spring 2017 – Spring 2021*

## RESEARCH FOCUS

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Hardware security architecture for post-quantum and privacy-preserving computation: protocol and system design, lattice-arithmetic datapath architecture (NTT, Keccak), side-channel and fault countermeasure integration, and silicon bring-up and security evaluation.

Two 28 nm cryptographic ASICs architected and taped out — a post-quantum authentication engine with on-chip TRNG and SCA/fault tolerance, measured in silicon and demonstrated live (CICC'26; JSSC under review), and a reconfigurable TFHE-CNN processor for encrypted inference with fault-injection characterization.

Hardware-security and ASIC research is complemented by a strong computer architecture and systems foundation, including graduate coursework and hands-on projects in processor microarchitecture, parallel computing, and system-level C/CUDA programming.

## FELLOWSHIPS & AWARDS

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**NAE Grand Challenge Vest Fellowship** (Security section, Nominated, MIT EECS)

Spring 2026

**MIT Presidential Graduate Fellowship Program**

Fall 2023 – Summer 2024

## RESEARCH EXPERIENCE

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### Post-Quantum Fuzzy Identity-Based Encryption (PQC-FIBE) Coprocessor with On-Device Biometric Key Derivation

*MIT, Research Assistant | DARPA*

*Spring 2026 – Present*

- Designing a unified cryptographic coprocessor that combines post-quantum security with personalized bio-authentication, eliminating reliance on external wireless communication for identity establishment in ingestible devices.
- Architecting a shared hardware datapath that multiplexes existing ML-KEM polynomial arithmetic (NTT, Keccak) for FIBE matrix operations, minimizing area/power overhead on ultra-constrained silicon.
- Developing a lightweight edge AI pipeline using multi-modal Siamese networks to extract stable “bio-fingerprints” from physiological signals, serving as identity keys for the PQC-FIBE scheme.

### Reconfigurable TFHE-CNN Processor for Encrypted Inference — 28 nm ASIC

*MIT, Research Assistant | ARPA-H*

*Fall 2025 – Present*

- Developed a 28 nm ASIC with a reconfigurable TFHE datapath at 0.16–0.33 mJ/PBS, performing encrypted biometric matching and inference directly on ciphertext with no raw-data exposure — deployed between hospital servers and body area networks (BANs). [S1]
- Characterized the encrypted-inference datapath under fault injection. Taped out May 2026.
- In IP patent process through MIT TLO [D2].

### Post-Quantum Authentication Engine with On-Chip TRNG and SCA/Fault Tolerance — 28 nm ASIC

*MIT, Research Assistant | ARPA-H*

*Fall 2023 – Spring 2025*

- Designed a 28 nm cryptoengine achieving  $9.8\text{--}36.6\times$  PQC energy gains at  $0.22\text{ mm}^2$  and  $0.86\text{ }\mu\text{J}/\text{Op}$ , with an on-chip  $8.5\text{ fJ/bit}$  TRNG and built-in side-channel and fault-injection countermeasures. [C1]
- Designed a lightweight post-quantum authentication protocol reducing energy by 61–76% on ARM Cortex-M4 through co-optimized PQC and lightweight cryptography (LWC); validated with in vitro/in vivo ingestible capsule prototypes. [C2]
- Performed chip bring-up and evaluation with Opal Kelly FPGA and custom PCB; live silicon demonstration at ISSCC’25 Student Research Preview.
- Extended version under review at *IEEE JSSC* (first author) [S2]; in IP patent process through MIT TLO [D1].
- Featured in MIT News [M1] and CNN [M2].

## Key-Switching Accelerator for CKKS Fully Homomorphic Encryption — 28 nm ASIC

Seoul National University, Research Assistant

Fall 2021 – Summer 2023

- Designed a 28 nm key-switching accelerator with top-to-bottom optimizations spanning algorithm, architecture, and external-memory traffic: 39% fewer external memory accesses,  $11.0\times$  faster than CPU, and  $12.7\times$  higher energy efficiency than prior works. [C4]

## High-Speed I/O and HBM Interface Research

Seoul National University, Research Assistant

Fall 2021 – Summer 2023

- SK Hynix collaboration: Developed WITCH crosstalk avoidance code for HBM, achieving 84.6% bit efficiency (+20.8% vs. prior) with 17–50% eye height improvement. [C3, J1]
- Samsung collaboration: Parametric Verilog sliding-block DFE for ADC-based SerDes receivers; BER simulation and optimization with XMODEL.

## Quantized LSTM Training and Inference in a GPU-Based Framework

Seoul National University, Internship

Summer 2021

- Implemented quantized LSTM support in a PyTorch/CUDA training framework, maintaining quantized numerical formats throughout both training and inference.
- Analyzed model weight and activation distributions to characterize the impact of quantization on model accuracy.

## PUBLICATIONS

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### Peer-Reviewed Conference Papers

[C1] **S. Jang**, S. Maji, R.S. Agrawal, H.S. Lee, G. Traverso, and A.P. Chandrakasan, “A 28nm  $0.86\mu\text{J}/\text{Op}$  Post-Quantum Secure Authentication Engine with  $8.5\text{ fJ/bit}$  TRNG and SCA/Fault Tolerance for Wireless Biomedical Devices,” *2026 IEEE Custom Integrated Circuits Conference (CICC)*. IEEE, 2026. (first author)

- Invited to submit extended version to *IEEE Solid-State Circuits Letters (SSC-L)*
- Invited to submit extended version to *Journal of Cryptographic Engineering (JCEN)*

[C2] **S. Jang**, H. S. Lee, R.S. Agrawal, S. Maji, et al., “Post Quantum Secure Communication Protocol with Ultra Low-Power Hardware Solution for Ingestible Medical Device,” *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2025. (first author)

[C3] **S. Jang\***, S. Jeon\*, K. Shin, D. Lee, et al., “WITCH: WeIghTed Coding Scheme for Crosstalk Reduction in High Bandwidth Memory,” *Proceedings of the 30th Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024. (\*equal first author)

[C4] **S. Jang**, S. Park, and D. Jeon, “Efficient Key Switching Accelerator for Fully Homomorphic Encryption,” *Proceedings of the 30th Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024. (first author)

### Peer-Reviewed Journal Papers

[J1] **S. Jang\***, S. Jeon\*, K. Shin, D. Lee, et al., “Weighted Coding Scheme for Noise Reduction in Silicon Interposer of HBM,” *IEEE Transactions on VLSI Systems (TVLSI)*, 2025. (\*equal first author)

### Manuscripts Under Review (not peer-reviewed)

[S2] **S. Jang**, R.S. Agrawal, S. Maji, G. Traverso, and A. P. Chandrakasan, “A  $0.86\text{-}\mu\text{J}/\text{Op}$  Post-Quantum Secure Authentication Engine with TRNG and SCA/Fault Tolerance for Wireless Biomedical Devices,” *IEEE Journal of Solid-State Circuits (JSSC)*. (Submitted, first author)

## INDUSTRY EXPERIENCE

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### Apple

Hardware Engineering Intern (*SerDes, ADC-based RX*)

Waltham, MA

Summer 2026

### Samsung Electronics

Industry Research Collaboration (*SerDes, ADC-based RX*) during M.S.

Seoul, South Korea

Fall 2021 – Summer 2023

### SK Hynix

Industry Research Collaboration (*HBM crosstalk reduction*) during M.S.

Seoul, South Korea

Fall 2021 – Summer 2023

## SKILLS

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**Security & Cryptography:** ML-KEM (Kyber), TFHE, CKKS, fuzzy identity-based encryption, lightweight cryptography (LWC); side-channel analysis and countermeasures, fault-injection tolerance, true random number generation

**Computer Architecture & Systems:** Graduate Advanced Computer Architecture: extended SimpleScalar for out-of-order execution and coarse-grained multithreading; Parallel Computing: distributed/parallel programming with CUDA and OpenMP; Systems Programming: C for system-level software

**Languages:** C, C++, CUDA, Python, Perl, GoLang, Verilog/SystemVerilog, Tcl, Matlab

**Design & Verification:** VCS/Xcelium, XMODEL, Design Compiler/Genus, ICC2, Innovus, PrimeTime, PrimePower, PrimeLib, Virtuoso

**Silicon Bring-up & Evaluation:** Opal Kelly FPGA, custom PCB design (Altium), ChipWhisperer

## PATENTS & INVENTION DISCLOSURES

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[D1] S. Jang, et al., “Area- and Energy-Efficient Post-Quantum Authentication Engine with On-Chip Entropy Source and Early-Termination Fault Detection,” MIT Technology Licensing Office, Case No. 27469, accepted August 2026. (*Under evaluation for patent filing*)

[D2] S. Jang, et al., “Reconfigurable TFHE-CNN Processor for Encrypted Personal-Hub Inference,” MIT Technology Licensing Office, Case No. 27502, accepted September 2026.

[P1] S. Jeon, **S. Jang**, K. Shin, H. Chi, W. Shin, C. Pyo, D. Jeon, “Semiconductor Device for Reducing Crosstalk Noise,” US Patent Application No. US20260031855A1, filed Oct. 2024, published Jan. 2026. Assignees: SK Hynix Inc., Seoul National University R&DB Foundation. (*Pending*)

[P2] S. Jeon, **S. Jang**, K. Shin, H. Chi, W. Shin, C. Pyo, D. Jeon, “Semiconductor Device for Reducing Crosstalk Noise,” US Patent Application No. US20260051924A1, filed Oct. 2024, published Feb. 2026. Assignees: Seoul National University R&DB Foundation, SK Hynix Inc. (*Pending*)

## PRESENTATIONS

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**CICC 2026 Oral Presentation** — 28nm post-quantum secure authentication engine [C1].

**ISSCC 2025 Student Research Preview (SRP)** — 28nm PQC cryptoengine with live silicon demo.

**EMBC 2025 Oral Presentation** — Ultra-low-power post-quantum secure communication protocol [C2].

**ARPA-H Sponsor Review Meetings** — Technical presentations and live demonstrations of PQC and TFHE hardware (recurring).

## MENTORING

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### Master of Engineering (MEng) Mentor

MIT

Summer 2026 – Present

- Supervising an MEng student extracting bio-fingerprints from physiological signals for integration into the PQC-FIBE on-device authentication scheme.

### Undergraduate Research Opportunities Program (UROP) Mentor

MIT

Spring 2026 – Summer 2026

- Supervised two first-year undergraduate researchers on differentiable logic gate networks (DLGNs) for TFHE-compatible neural network inference.

## PRESS/MEDIA COVERAGE

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[M1] “New chip can protect wireless biomedical devices from quantum attacks”, *MIT News*, April 23, 2026.

[M2] “Q-Day is almost here. It could unleash a cybersecurity crisis far worse than Y2K”, *CNN*, May 17, 2026.